NATIONAL TEACHERS COLLEGE

School of Arts, Sciences and Technology
Bachelor of Science in Information Technology

DISCUSSION DOCUMENT

ISO/IEC 25010 Metrics for Evaluation

For Section 4.5 (UAT and Evaluation) and Chapter 5, Section 5.2 (Evaluation Results)

1. What Is ISO/IEC 25010?

ISO/IEC 25010 is an international standard that defines **what software quality means**. It is part of the ISO/IEC 25000 family, known as SQuaRE (Systems and Software Quality Requirements and Evaluation). The standard gives a common list of **quality characteristics** and **sub-characteristics**. Your capstone uses this list to decide what to evaluate, so that you do not simply ask users, "Do you like the system?"

The standard tells you **what** to evaluate. It does not fix the questions you must ask or the scale you must use. Your team chooses the **metrics**, meaning the ways you will measure each characteristic, and explains them in Chapter 4.

Which edition should we use?

ISO/IEC 25010 has two editions. The **2011 edition** has **eight** characteristics and is the one used in most existing capstone papers. The **2023 edition** has **nine** characteristics. It replaces Usability with *Interaction Capability*, replaces Portability with *Flexibility*, and adds *Safety*. **Ask your adviser or panel which edition your department follows.** Whichever you choose, use it consistently in your questionnaire, tables, and discussion, and cite the standard in APA format.

2. The Quality Characteristics

Table 1 shows the eight characteristics of the 2011 edition, their sub-characteristics, and the plain question each one answers.

Table 1. ISO/IEC 25010:2011 Product Quality Characteristics

Characteristic	Sub-characteristics	What it asks
Functional Suitability	Functional completeness, functional correctness, functional appropriateness	Does the system do what it is supposed to do, and do it correctly?
Performance Efficiency	Time behaviour, resource utilization, capacity	Is it fast enough, and does it use resources well?
Compatibility	Co-existence, interoperability	Can it work with other systems, browsers, and devices?
Usability	Appropriateness recognizability, learnability, operability, user error protection, user interface aesthetics, accessibility	Can users learn and use it easily, with few errors?
Reliability	Maturity, availability, fault tolerance, recoverability	Does it work when needed and recover from failures?
Security	Confidentiality, integrity, non-repudiation, accountability, authenticity	Is data protected from unauthorized access and changes?
Maintainability	Modularity, reusability, analysability, modifiability, testability	Can developers fix or change it easily?

Characteristic	Sub-characteristics	What it asks
Portability	Adaptability, installability, replaceability	Can it be moved to and installed in other environments?

Table 2 summarizes what changed in the 2023 edition.

Table 2. Main Changes in ISO/IEC 25010:2023

Change	Details
Nine characteristics	The model now has nine product quality characteristics instead of eight.
Usability becomes **Interaction Capability**	Adds sub-characteristics such as inclusivity and self-descriptiveness.
Portability becomes **Flexibility**	Adds scalability as a sub-characteristic.
Safety is new	Sub-characteristics: operational constraint, risk identification, fail safe, hazard warning, and safe integration.
Other sub-characteristic changes	Reliability uses faultlessness in place of maturity, and Security gains resistance.
Quality in use	The separate quality-in-use model of the 2011 edition (effectiveness, efficiency, satisfaction, freedom from risk, context coverage) moved to ISO/IEC 25019.

Not every characteristic needs equal weight. Choose the ones that fit your system and explain why in Chapter 4. Many capstones evaluate all of them, but a system with no safety risk, for example, may reasonably leave Safety out if the 2023 edition is used. Your panel may ask you to justify any characteristic you leave out.

3. What Are "Metrics" in a Capstone Evaluation?

A **metric** is a defined way of measuring a quality characteristic. Most capstone papers use two kinds, and the strongest papers use both.

Table 3. Two Kinds of Evaluation Metrics

	Perception-based (Survey)	Measurement-based (Objective)
How it works	Users and experts rate statements on a scale, usually a 5-point Likert scale.	You measure something with a tool or a count: seconds, percentages, number of failures.
Result	Weighted mean and verbal interpretation.	A number compared with a target, such as "3 seconds or less".
Strength	Captures the users' real experience and satisfaction.	Not affected by opinion; can be repeated.
Weakness	Can be biased or influenced by the demonstration.	Does not show how users feel.
Where reported	Sections 4.5 (method) and 5.2 (results)	Section 4.4 (test results) and Section 5.2 or 5.3

The ISO/IEC 25000 family also has a part, ISO/IEC 25023, that defines measures for product quality. You may look at it for ideas, but you are not required to apply every measure. What matters is that each metric you choose is **clear, measurable, and suitable for your system**.

4. Sample Metrics per Characteristic

Table 4 gives sample questionnaire statements and sample objective measures for each characteristic. Adapt them to your own system. Use your own numbers and targets.

Table 4. Sample Metrics for Each Characteristic

Characteristic	Sample Survey Statement	Sample Objective Measure
Functional Suitability	The system has all the functions I need for my tasks.	Functional completeness = 1 − (A ÷ B), where A is the number of specified functions that are missing and B is the number of functions specified.
Performance Efficiency	The system responds quickly to my actions.	Average response time in seconds with a set number of users, compared with a target (for example, 3 seconds or less).
Compatibility	The system works well on the browsers and devices I use.	Compatibility rate = (browsers/devices passed ÷ browsers/devices tested) × 100.
Usability (Interaction Capability)	The system is easy to learn and use.	Task completion rate = (tasks completed successfully ÷ tasks attempted) × 100; average time per task.
Reliability	The system works without failing when I need it.	Availability = operating time ÷ (operating time + downtime) × 100; number of failures during the test period.
Security	The system keeps my data protected from other people.	Security test pass rate = (security tests passed ÷ security tests run) × 100.
Maintainability	The system is organized so that it can be updated easily.	Test coverage = (functions covered by tests ÷ total functions) × 100.
Portability (Flexibility)	The system can be installed and used in a different environment.	Installation success rate = (successful installations ÷ attempted installations) × 100.
Safety (2023 only)	The system prevents actions that could cause harm or loss.	Percentage of critical actions (such as delete) protected by a confirmation or warning.

5. Building the Questionnaire

- **Group the statements by characteristic.** Write about **five statements** for each one, based on its sub-characteristics.
- **One idea per statement.** "The system is fast and easy to use" is two ideas. Split it.
- **Write positive, clear statements** in simple language that your respondents understand.
- **Use a 5-point Likert scale** so that the same scale is used throughout.
- **Validate the instrument.** Have your adviser and at least two experts review it before use. Record their comments and changes.
- **Test its reliability.** Give it to a small pilot group first. A Cronbach's alpha of about 0.70 or higher is commonly accepted as good internal consistency. Tools such as SPSS, Excel add-ins, or Python can compute it. Ask your adviser what your department requires.
- **Attach the final questionnaire** as Appendix C.

6. Rating Scale and Verbal Interpretation

The mean ranges come from dividing the scale into equal parts. For a 5-point scale, the interval is (5 − 1) ÷ 5 = **0.80**. Starting from 1.00, each range is 0.80 wide (rounded to 0.01 in the table).

Table 5. Sample Rating Scale

Scale	Mean Range	Verbal Interpretation
5	4.21 to 5.00	Strongly Agree (Excellent)
4	3.41 to 4.20	Agree (Very Good)
3	2.61 to 3.40	Neutral (Good)
2	1.81 to 2.60	Disagree (Fair)
1	1.00 to 1.80	Strongly Disagree (Poor)

Use the **same scale and wording** in Chapter 4, Chapter 5, and Appendix C.

7. Computing the Results

The usual statistical treatment is the **weighted mean**:

$$\mathbf{WM = \Sigma(f \times w) \div N}$$

where **f** is the number of respondents who gave a rating, **w** is the rating value, and **N** is the total number of respondents.

Worked example (made-up data)

For the statement "The system is easy to learn," 10 respondents gave these ratings:

Table 6. Sample Computation of the Weighted Mean

Rating (w)	Number of Respondents (f)	f × w
5	6	30
4	3	12
3	1	3
2	0	0
1	0	0
Total	**N = 10**	**45**

WM = 45 ÷ 10 = **4.50**, which is interpreted as **Strongly Agree (Excellent)** using Table 5.

- The mean of a characteristic is the average of the means of its statements.
- The overall mean is the average of the characteristic means.
- If you have different groups (for example, users and IT experts), report the mean **per group and overall**, and use the same scale for all.

8. Choosing the Respondents

- **End users** are best placed to rate Functional Suitability, Usability (Interaction Capability), and Performance Efficiency because they use the system directly.
- **IT experts** are best placed to rate Security, Maintainability, Compatibility, Portability (Flexibility), and Reliability, which need technical knowledge.
- Explain **how** respondents were chosen (for example, purposive sampling) and why they qualify.
- Ask your adviser or panel about the required number of respondents. Some teams use a formula such as Slovin's when the population is known, while others follow a minimum group size set by the department. Whatever you use, state it and justify it.

9. Presenting and Interpreting the Results

- **Section 4.5** explains the method: respondents, instrument, scale, procedure, and treatment.
- **Section 5.2** presents the mean scores per characteristic in a table, with the verbal interpretation and, if helpful, the rank. Explain each table right after it.
- **Section 5.3** explains what the results mean: the highest and lowest characteristics and why, how they relate to your test results in Section 4.4, your objectives, and the studies in Chapter 2, and the limitations of your study.
- Report the **lowest scores honestly** and suggest how to improve them. Panels trust honest results more than perfect ones.

10. Common Mistakes to Avoid

- Using the ISO/IEC 25010 name but a questionnaire that does not follow its characteristics.
- Mixing the 2011 and 2023 characteristic names in the same paper.
- Different scales or mean ranges in Chapter 4, Chapter 5, and the appendix.
- Statements that ask two things at once or use leading words such as "excellent".
- Rating technical characteristics only with students who cannot judge them.
- Vague targets ("fast") instead of measurable ones ("3 seconds or less").
- Showing a table of scores with no explanation or comparison with the literature.
- Presenting numbers that were not actually collected. Keep the raw data and questionnaires.

11. Discussion Questions

1. Which edition of ISO/IEC 25010 will your team use, and why?
2. Which two or three characteristics matter most for your system? Which would you rate with objective measures, and which with a survey?
3. Write one good and one poor survey statement for Security. What makes the poor one poor?
4. If your Usability score is high but your Reliability score is low, what would you explain in Section 5.3, and what would you recommend?
5. Who would be the best respondents for each characteristic in your project?

Reference to cite: International Organization for Standardization. (2023). *Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model* (ISO/IEC 25010:2023). https://www.iso.org/standard/78176.html